PHYS30762 Object Oriented Programming in C++

Analogue Circuits Report

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Abstract

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1. Introduction

This program allows users to build analogue circuits using resistors, capacitors, and inductors (which are all classes derived from an abstract base `Component` class). A `Circuit` class can be set to be series or parallel, and any number of components can be added to it, which is then stores as a vector of components (specifically a vector of unique pointers to components). Since the `Circuit` class is inherited from the `Component` class itself, it is possible to add series circuits to parallel circuits, or vice versa, and therefore construct circuits of any topology.

In the main file of this program, a *library* of components is created, which stores different types of resistors, capacitors and inductors. These components are validated through the use of validation files, such that if the user tries to initialise a component with properties which do not match those specified in the validation file, an error message will be returned. The validation files specify a component name, like *Wirewound* for resistor, as well as a tolerance to dissipative effects, and a maximum frequency to which they can withstand [5]. Each validation file then contains maximum and minimum possible values for the resistance, capacitance and inductance respectively.

The `Component` base class contains an operating frequency `frequency` and a unique ID `component_id`, as well as a function to calculate the impedance of the component (using the complex library in C++), and its magnitude and phase difference (`get_impedance()`, `get_impedance_magnitude`, `get_phase_difference`). It also has a `base_info()` function which outputs these properties, and which is overriden in the derived classes to print the resistance, capacitance and inductance of the corresponding classes.

2. Code design and implementation

Due to the use of a vector of unique pointers to store the components in the `Circuit` class, adding components to circuits or moving them is not trivial. The process is best-described using an analogy, considering the user of the program to be a student in an undergraduate circuits lab. This student wants to build a circuit using components from the lab's demonstrator, who has this library of components available. This library must contain components which are manufacturable, and the validation files could be seen as catalogues of different types of each component which can be manufactured.

The student wants to use one of the components from the library of the lab demonstrator, but they cannot take it directly away from the lab demonstrator, as the demonstrator is the owner of that specific component. If they let the student take that one component which they own, then no one else in the lab could use it. In that sense it is not logical for identical components to be stored twice in the library. The student must make a copy of the component from the library, at which point they can add it to their circuit. This is done via a `clone()` function which all of the `Component`-derived classes have. They could initialise a component just by itself and then move it to the circuit, but its easier to just clone inside the circuit infrastructure.

3. Results

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4. Discussion and conclusions

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